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IMPACT OF INCOME MAINTENANCE ON LOW BIRTH WEIGHT: EVIDENCE FROM THE GARY EXPERIMENT*

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ABSTRACT

Birth weight is a useful index of infant health. Low birth weight (5.5 pounds or less) is associated with high rates of infant mortality and morbidity. Low birth weight is also associated with low socioeconomic status. The question arises, therefore, whether income transfers can affect the incidence of low birth weight among the poor. The impact of an expanded income support plan on low birth weight was analyzed using data on 404 infants born to participants in the Gary Income Maintenance Experiment. A significant health response was observed for children of women who face high-risk pregnancies.

An infant's weight at birth is an important index of its health status. Low birth weight, defined as 2500 grams (5.5 pounds) or less, is associated with a sharply elevated risk of death in the first year of life and with higher rates of morbidity both during infancy and later years. It is a commonplace of the public health literature that low birth weight is a significant problem in the United States, particularly among blacks and the economically disadvantaged.

Given that a number of factors identified as having causal impact on birth weight are related to income or to socioeconomic status, the question

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arises whether income transfers can affect the incidence of low birth weight among the poor. This paper presents findings from a study of the impact of income maintenance on the incidence of low birth weight in an urban, black, low-income population. The analysis is based upon data on 404 single live births to participants in the Gary Income Maintenance Experiment.

The remainder of this paper is organized as follows: Section I contains a brief description of the Gary Income Maintenance Experiment. In order to motivate the discussion of the impact of income maintenance on low birth weight, in Section II we review data on the incidence of low birth weight and on variations among demographic and socioeconomic groups, and summarize the etiology (or, causes) of low birth weight. We also present our hypotheses regarding the mechanisms by which income maintenance might intervene in that etiology. In Section III we develop theoretical and econometric models of birth-weight response to income maintenance. Findings are presented in Section IV, and Section V contains concluding remarks.

I. THE GARY INCOME MAINTENANCE EXPERIMENT

The income maintenance experiment conducted in Gary, Indiana, from 1970 to 1974 was a federally sponsored test of the negative income tax (NIT) as an alternative to traditional forms of public welfare. All participating families were black and among them was a high concentration of female-headed households.[1] In this section, we provide a capsule description of the features of the Gary experiment as background for the discussion of the present study. Details on the administration and operations of the experiment are provided elsewhere (Kehrer et al. [17]).

A negative income tax plan subsidizes a family's income in accordance with the basic principles contained in the simple payment formula:

$$P = \begin{cases} G - tE & \text{if } (G - tE) > 0 \\ 0 & \text{if } (G - tE) \leq 0 \end{cases}$$

where P is the payment, G is the guarantee level, t is the tax rate, and E is earnings. Moreover, in Gary, uninsured medical expenses above a fixed deductible ($30 per family member over a six-month period) were subtracted from family earnings before applying the payment formula.

1 Similar experiments were conducted in New Jersey and Pennsylvania, in Seattle and Denver, and in rural areas of North Carolina and Iowa, as part of a broad federal effort to explore the influences of an NIT on work and job search efforts, expenditures, attitudes, and a wide variety of other economic and social behaviors. Each of the four experiments focused on a sample with different demographic characteristics.

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The experimental design in Gary included two guarantee levels (given family size) and two tax rates. The guarantee levels were $4300 (about equal to the 1971 poverty line) and $3300 per year for a family of four. The guarantees were varied with family size to reflect economies of scale (for example, a family of two received two-thirds as much as a family of four under either plan) and were adjusted semiannually to compensate for increases in the cost of living. The two alternative tax rates were 40 percent and 60 percent. The breakdown level of earnings, at which payments became zero, is given by G/t. For the least generous plan in Gary (G = $3300, t = 60 percent), it was $5500; for the most generous plan (G = $4300, t = 40 percent), it was $10,750.

Experiment participants were drawn from the residents of low-income neighborhoods in Gary. Based on their pre-experiment income and demographic characteristics, eligible families[2] were assigned randomly to one of the four treatment plans or to a control group.[3] Table 1 presents the basic sample design of the Gary experiment. Families are differentiated by type (husband-wife or female-head) and by experimental plan. Of the 1799 families participating in the plan at enrollment, 1028 were eligible to receive NIT payments and 771 were controls,[4] who were not eligible to receive payments. Among the former group, families received twice-monthly payments when their previous month's earnings (after allowable deductions) were below the breakeven level.

II. THE LOW-BIRTH-WEIGHT PROBLEM

Table 2 gives the distributions of birth weights by race for the 1973 United States live birth cohort. Of all infants born in 1973, 7.6 percent weighed less than 2501 grams at birth. However, substantial differences in the incidence of low birth weight are exhibited between the races: the rate for black infants (13.3 percent) was more than twice that for whites (6.4 percent). Moreover, the relative disparity between black and white rates seems to have been

2 The eligibility criteria were that the household head was black and that at least one dependent under age 18 was present. For details on sample selection and enrollment, see Kehrer et al. [17], pp. 12–19.

3 In the Gary experiment as in the other income maintenance experiments, the Watts-Conlisk allocation model was used as the basis of the sample design. A feature of this model is that treatment status (i.e., treatment or control) is not purely random, but only random conditional on characteristics of families in the sample.

4 It is important to be aware that many control families received benefits under a variety of public assistance programs, including Aid to Families with Dependent Children. For families eligible to receive experimental NIT payments, such benefits were subject to a 100 percent tax. However, experimental treatment was generally better than alternative forms of welfare.

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TABLE 1
GARY INCOME MAINTENANCE EXPERIMENT SAMPLE DESIGN BY TREATMENT STATUS AND FAMILY TYPE

Treatment Status	Family Type		
	Husband-Wife	Female-Headed	All Families
Experimental treatment families			
Guarantee = $4300			
Tax rate = 40%	81	122	203
Tax rate = 60%	76	122	198
Guarantee = $3300			
Tax rate = 40%	136	177	313
Tax rate = 60%	110	204	314
All NIT families	403	625	1028
Control families	326	445	771
All families	729	1070	1799

TABLE 2
PERCENTAGE DISTRIBUTION OF LIVE BIRTHS BY BIRTH WEIGHT AND RACE: UNITED STATES, 1973

Birth Weight (grams)	Race		
	All Races[a]	White	Black
2500 or less	7.6	6.4	13.3
2501 or more	92.4	93.6	86.7
1000 or less	0.5	0.4	1.1
1001–1500	0.6	0.5	1.2
1501–2000	1.4	1.2	2.6
2001–2500	5.0	4.3	8.4
2501–3000	18.2	16.6	25.8
3001–3500	38.2	38.1	38.4
3501–4000	27.0	28.8	18.2
4001–4500	7.5	8.3	3.7
4501–5000	1.3	1.5	0.6
5001 or more	0.2	0.2	0.1

Source: Calculated from frequency distributions given in National Center for Health Statistics [32].
a There was a total of 3,136,965 births in 1973, of which 2,551,030 were classified as white and 512,597 were classified as black (Negro). The rates of this table are based on the 3,128,318 births of all races, 2,544,166 white births, and 511,022 black births for which birth weight was recorded.

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TABLE 3
INFANT MORTALITY RATES PER 1000 LIVE BIRTHS, BY BIRTH WEIGHT, COLOR, AND SEX: UNITED STATES, 1960 LIVE-BIRTH COHORT

	Total			White			All Other		
Birth Weight (grams)	Both Sexes	Male	Female	Both Sexes	Male	Female	Both Sexes	Male	Female
All birth weights	25.1	28.3	21.8	22.2	25.1	19.1	41.4	46.6	36.7
1000 or less	919.3	935.3	901.9	929.3	945.5	911.3	893.6	908.2	878.7
1001–1500	548.4	613.0	478.1	575.6	634.3	508.3	478.1	552.6	406.4
1501–2000	206.6	254.4	160.7	219.0	268.3	170.3	171.8	213.2	135.1
2001–2500	58.4	74.6	45.1	58.2	75.3	43.9	59.4	72.8	48.7
2501–3000	19.0	24.0	15.1	17.3	22.1	13.6	25.3	31.0	20.6
3001–3500	10.1	12.1	8.3	8.9	10.6	7.2	17.4	20.1	14.7
3501–4000	7.8	8.9	6.8	6.9	7.7	5.8	17.0	18.5	15.2
4001–5000	8.3	8.8	7.6	7.0	7.4	6.3	21.0	22.1	19.1
4501 or more	13.3	13.4	12.5	11.2	11.4	10.9	28.1	29.8	25.3

Source: National Center for Health Statistics [30], pp. 30–32.

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increasing, with blacks becoming relatively worse off over time.[5] The implications of these differences are serious, since high rates of low birth weight portend higher rates of infant mortality and morbidity.

Implications of Low Birth Weight

Infant mortality rates for the 1960 United States live birth cohort are given in Table 3 by birth weight, race, and sex. Low birth weight has a strong association with infant mortality. According to the National Center for Health Statistics (NCHS) [28], "Birth weight is one of the most important predictors of risk of death in infancy." Chase [5] notes that, among the 1960 U.S. live birth cohort, the infant mortality rate was 17 times greater for birth weights of 2500 grams or less than for birth weights greater than 2500 grams (190.3 deaths per thousand live births compared with 11.2 per thousand).[6]

Disparities in infant death rates among racial groups in the United States (see, for example, Hunt [12]) and unfavorable comparisons between infant mortality rates in the United States and in other countries (see Wallace and Goldstein [47]) both suggest that the overall domestic rate is higher than it need be, given the state of medical knowledge. Moreover, when such comparisons are drawn, differences in the incidence of low birth weight often are cited as a possible source of the different rates of infant mortality (see, for example, Fuchs [11], Naeye et al. [26], and NCHS [29]).

In addition to higher rates of infant mortality, those low-birth-weight infants who survive their first year tend to experience a wide variety of health problems compared with normal-birth-weight children. Research on low birth weight and subsequent morbidity has suggested that there is an association between low birth weight and a greater risk of physical problems (for example, see Douglas and Mogford [8], Knobloch et al. [21], Drillien [9], and Scarr [41]). However, it is not always clear whether low birth weight is cause or effect:

> For some [conditions associated with low birth weight], low birth weight may

5 Data published by the National Center for Health Statistics (NCHS) suggest that the overall incidence of low birth weight has been fairly constant since 1950, with rates of 7.5 percent in 1950 and 7.7 percent in 1960 [31] similar to the 7.6 percent for 1973 shown in Table 2. Rates for nonwhites compared with whites show increasing relative differences since 1950: rates were 10.2 percent for nonwhites and 7.1 percent for whites in 1960 (the nonwhite rate 1.4 times the white rate), and 12.8 percent for nonwhites and 6.8 percent for whites in 1960 (the nonwhite rate 1.9 times the white rate). The black rate of 13.3 shown in Table 2 for 1973 was 2.1 times the rate of 6.4 for whites. However, since the earlier data do not separate out blacks from all nonwhites, they are not strictly comparable to the 1973 rates.

6 Similarly, Kessner et al. [19] found that, among all live births in New York City in 1968, the infant mortality rate for low-birth-weight children (140.5 per 1000) was 17 times that for infants whose birth weights were greater than 2500 grams (8.4 per 1000).

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> be a consequence of the associated condition rather than a correlate of the causative mechanism. For example, an infant with severe congenital anomalies of the musculoskeletal system may be of low birth weight because the fetus was unable to grow to a fully developed infant after it was affected in utero. . . . On the other hand, other conditions may be resultants of low birth weight . . . in the etiologic sense. [31, p. 2]

Etiology of Low Birth Weight and the Impact of Income Maintenance

In order to establish a rationale for expecting an income maintenance plan to affect the incidence of low birth weight, it is necessary to show that some of the causes of low birth weight are subject to change. Furthermore, those causes that are not subject to change must be identified and controlled for in the analysis.

Factors associated with high risk of low birth weight fall into three general categories: (1) demographic characteristics of the mother and infant, (2) physiological characteristics of the mother, and (3) behavioral practices of the mother. Table 4, which contains a summary list of these factors, is provided as a reference. The discussion in the text is confined to those etiological influences on low birth weight which are expected to be amenable to intervention by the NIT.

Among the factors identified in Table 4 are a number of physiological conditions which, by virtue of being diagnosable and treatable, are likely to respond to prenatal care.[7] To the extent that utilization of prenatal care tends to increase in response to an NIT, the incidence of low birth weight also may be expected to exhibit a response. There are three reasons for expecting an income maintenance program to encourage increased utilization of prenatal care. First, by increasing income, the program increases ability to pay for medical care. Second, the program effectively subsidizes the time cost involved in obtaining medical services for an individual who must take unpaid time off from work to visit the doctor. (The tax rate feature of the NIT provides partial compensation for the loss in earnings resulting from the loss in work time.) Finally, the Gary plan subsidy of the money cost of medical services received is likely to further encourage early and adequate use of prenatal care.

Two further, and possibly related, causes of low birth weight which may be responsive to income maintenance are maternal malnutrition and limited maternal weight gain during pregnancy. The impact of the NIT on these two factors may derive either from a direct effect of increased family income on

7 More generally, a number of studies described in the public health literature (e.g., Weiner and Milton [48], Iba et al. [14], Kessner et al. [19], and Zackler et al. [50]) point to the importance of the timing and quantity of prenatal care in reducing the probability of low birth weight.

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the quality of the family diet or from an indirect effect, through increased utilization of prenatal care.[8]

Finally, some limited evidence (Stewart [44], and Kauppinen [15]) suggests that the risk of low birth weight is higher if a woman worked at paid employment during pregnancy as compared with women who do only housework in their own homes. Given that an income maintenance plan provides an opportunity for individuals to substitute subsidized leisure time for market work, the reduced labor force activity permitted by the NIT may also tend to have a beneficial impact on birth weight.

III. A MODEL OF BIRTH-WEIGHT RESPONSE

We view the impact of income maintenance on birth weight as the outcome of a set of family responses. The model underlying our analytical strategy is a simultaneous-equations system in which birth weight reflects income maintenance effects on family income and on hours worked. In this section, we describe the measure of birth-weight response used and set forth the structural and reduced-form models that underlie our empirical work, and the econometric specification that was estimated.

Birth-Weight-Outcome Measure

The measure of birth-weight response, R, in our analytical formulation focuses on change in the left-hand tail of the birth-weight distribution. Since a wide range of birth weights are considered normal and healthy, there is no reason to expect that a health response to the NIT would take the form of moving an infant from one point within the healthy range to another point within that range. Moreover, if this *were* the impact of income maintenance, it could not be considered a significant health effect. Instead, it was felt that the most important impact would be to shift an infant from some point in the low-birth-weight range to a point closer to or within the healthy birth-weight range. The birth-weight response measure was defined as:

$$R = \begin{cases} (\alpha - BW) & \text{if } BW < \alpha \\ 0 & \text{if } BW \geq \alpha \end{cases} \tag{1}$$

where BW is actual birth weight and α is a critical value signifying the minimum of the healthy birth-weight range. The focus thus is on the difference between actual birth weight and a healthy birth weight. Larger

8 Rush, Davis, and Susser [39] argue that improvements in nutrition in a black urban population in the U.S. may be reflected in increased maternal weight gain during pregnancy and thereby in increased birth weight.

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TABLE 4
SOME FACTORS ASSOCIATED WITH HIGH RISK OF LOW BIRTH WEIGHT FROM A REVIEW OF THE LITERATURE

Category	Factor
Demographic	Maternal age under 18 and over 35[a]
	Birth order: first births and orders greater than two[a,f]
	Infant's sex female[a]
	Plurality (i.e., multiple births)[a,b]
	Nonwhite race[a,c]
	Maternal residence at high altitude[d]
Physiological	Low maternal prepregnancy weight[a]
	Intrauterine infections, such as rubella, cytomegalic inclusion disease, toxoplasmosis,[e] and syphilis[e]
	Chromosomal abnormalities
	Maternal cyanotic heart disease
	Placental insufficiency
	Asymptomatic bacteriuria[e]
	Maternal pre-eclampsia[e]
	Maternal eclampsia, or toxemia[e]
	Maternal chronic hypertension[e]
Behavioral	Short interval between pregnancies[a]
	Maternal cigarette smoking
	Heavy maternal alcohol consumption
	Maternal drug addiction
	Limited maternal weight gain during pregnancy
	Maternal malnutrition
	Work at paid employment during pregnancy

Source: Babson and Benson [1], Bergner and Susser [3], Clarren and Smith [6], Donnelly [7], Eastman and Jackson [10], Hutchison [13], Kauppinen [15], Krause [22], McDonald [23], Meyer [24], National Center for Health Statistics [31], Nesbitt et al. [33], Niswander et al. [34], Penchaszadeh et al. [37], Schaffer and Avery [42], Selvin and Janerich [43], Stewart [44], Sweet [45], and Wray [49].

a Not subject to change once conception has taken place.
b Not relevant to present study: only single births were included.
c Not relevant to present study: all observations are black.
d Not relevant to present study: all observations are in Gary or vicinity.
e Medical or obstetric conditions which are diagnosable and treatable.
f Effect of birth order varies with mother's age.

values of the dependent variable signify lower birth weight. Values of zero occur when an infant's birth weight is greater than or equal to α.

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The Structural Model

The structural model of birth-weight response to income maintenance is derived from the public health literature on the causes of low birth weight and from neoclassical labor supply theory:

$$E(R) = f(Z, Y, N, H_m) \tag{2}$$

$$Y \equiv \Sigma J_i + Y_n + P \tag{3}$$

$$\Sigma J_i \equiv \Sigma w_i H_i \tag{4}$$

$$P \equiv G(N) - t(\Sigma J_i + Y_n) \tag{5}$$

$$E(H_i) = g(W, H, Y_n, t, G(N)) \qquad i = 1, \ldots, k \tag{6}$$

where R = birth-weight response measure; Z = a vector of demographic, physiological, behavioral, and socioeconomic characteristics which influence birth weight; Y = current family income; N = family size; H_m = hours worked per week by the mother during pregnancy; J_i = current earnings per period of the ith family member; ΣJ_i = family earnings; Y_n = nonwage income other than the NIT payment; P = transfer payment received from the income maintenance plan; H_i = hours worked per period by the ith family member ($i = m$ for expectant mother); H = a vector of hours worked by family members other than the ith family member; w_i = wage of ith family member; W = a vector of family wage rates; t = NIT tax rate; and G = NIT guarantee level.

The model given by equations (2) through (6) consists of three definitions [(3), (4), and (5)] and $k + 1$ structural equations [(2) and the set of equations (6)]. The three definitions—current family income, family earnings, and the NIT payment—should be self-explanatory. Equation (2) is based on a review of medical and public health literature and on the hypotheses regarding birth-weight response to the NIT, summarized above. It should be noted that the vector Z includes not only the etiological characteristics reported in Table 4, but also indirect influences on birth weight such as mother's education and family health insurance coverage. These variables are expected to affect health behavior during pregnancy, including utilization of prenatal care, appropriate diet, sufficient rest, and so on.

Equations (6) are a set of labor supply equations for the family. These are based on a neoclassical labor supply model of individuals who take account of family income and labor supply of other family members in their own labor supply decision (see Kerachsky and Mallar [18]). The set of labor supply equations would contain one equation for each household member whose earnings might be used to support the expectant mother.

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The Reduced Form

Our interest in not with issues in labor supply responses, however, but with the response of birth weight to changes in family income and hours worked under an NIT plan. Consequently, we focus attention on the reduced-form equation:

$$E(R) = h(Z^*, Y_n, N, W, t, G) \tag{7}$$

where Z^* are those exogenous influences on birth weight contained in Z. The estimated reduced-form parameters enable a straightforward assessment of the overall impact of the NIT on birth weight. It is the reduced-form coefficients that are the parameters of the stochastic process that generated the observed sample. As such, the reduced-form parameters represent the interaction of underlying structural phenomena. Estimated coefficients of the NIT parameters allow convenient calculation of "what the experiment did" for the incidence of low birth weight.

Econometric Specification

Equation (7) is the basis for our econometric model of birth-weight response. In this section, we describe the variables used to estimate the econometric model.

Dependent Variable. The measure of birth-weight response given in equation (1) defines a limited dependent variable with a lower limit of zero. Hence, equation (7) may properly be estimated using Tobin's [46] maximum likelihood method (Tobit).

Ceteris paribus, the proportion of observations taking on the limit value will be determined by the value assigned to α, the minimum of the healthy birth-weight range. The literature on low birth weight suggests that α should be defined as 2501 grams (5.5 pounds). However, had this definition been employed here, 88.1 percent of the observations would have taken on the limit value of zero. The ability to estimate parameters in a Tobit model depends in part on the proportion of limit observations in the study sample. In cases where the sample has a high proportion of limit observations, estimated parameters have large variances, ceteris paribus. In an extreme case, when all observations are at the limit, it is not possible to estimate the model. Inasmuch as infant mortality is at a minimum for nonwhites weighing 3001 to 4000 grams at birth (see Table 3), α was set equal to 3001 grams (6.6 pounds). To the extent that 3001 grams may be viewed as the low end of an "optimal" range of birth weights, our findings should reflect experimental effects in moving infants from less-than-optimal birth weights into the optimal range.

Explanatory Variables. Two broad classes of explanatory variables are

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included in the econometric model. The first class consists of exogenous variables in the theoretical structure of birth-weight response, modified where necessary by limitations in the data set. The second is comprised of characteristics intended to account for systematic variations across families in preference structures underlying the labor supply equations and in opportunity sets. Table 5 provides definitions and a summary listing of the explanatory variables used.

The exogenous variables in the process underlying birth-weight response are given on the right-hand side of equation (7): Z^*, Y_n, N, W, t, G. The first nine variables given in Table 5 are elements of the vector Z^*—that is, they are the exogenous demographic, physiological, behavorial, and socioeconomic factors that influence birth weight. They include whether the child is male (*MALE*) and whether the mother's age was less than 18 (*AGELT*18) or greater than 34 (*AGEGT*34); the number of previous live births to the mother (*PARITY*) and whether the infant was the mother's first (*PRIMPARA*);[9] whether the child's mother had a previous delivery (live birth or fetal birth) less than 16 months prior to the current birth (*INTLT*16);[10] whether the mother is not a high school graduate (*EDLT*12); and whether the mother indicated in an experimental survey interview that she smokes (*SMOKER*). Ideally, Z^* also would include (a) variables which measure health knowledge and practices during pregnancy, and (b) indicators of existing health conditions associated with low birth weight which changes in income are not likely to change (e.g., cyanotic heart disease, rubella during pregnancy). Limitations of the data set, however, precluded inclusion of these variables. The mother's rate of successful previous pregnancies (*SUCCESS*) is used as a proxy for the presence of maternal health conditions likely to result in an infant of low birth weight despite the level of income.[11]

Inclusion of per capita nonwage family income (*NEARNPC*) and the mother's predicted market wage rate (*PWAGEMOM*) among the explanatory variables follows directly from the theoretical model of birth-weight response to an NIT. Per capita pre-enrollment earnings of other family members (*ERNOTHPC*) is used as a measure of wage rates of household members other than the mother.[12]

9 The public health literature suggests that the influence of parity on birth weight is not smooth. Parity zero seems to have a larger influence on average birth weight than do higher order parities (see, for example, Rush et al. [39], p. 376).

10 Similar results were obtained when this variable, which might perhaps be endogenous, was not included.

11 Data from a study of live births in the United States during January–March 1950 indicated that the incidence of low birth weight was more than 1.5 times as high for women who had reported at least one prior fetal death as for other women. See [27], p. 24.

12 An alternative to using *ERNOTHPC* is to use a predicted wage rate for each "relevant" household member. This procedure is complicated by the difficulties inherent in establishing

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TABLE 5
EXPLANATORY VARIABLES USED TO ESTIMATE
THE REDUCED-FORM BIRTH-WEIGHT EQUATION

Z^*: Social, demographic, behavioral, and physiological characteristics		
MALE	≡	1 if child is male; 0 if child is female
*AGELT*18	≡	1 if mother was younger than 18 at time of child's birth; 0 otherwise
*AGEGT*34	≡	1 if mother was more than 34 years old at the time of child's birth; 0 otherwise
PARITY	≡	Number of previous live births to mother
PRIMPARA	≡	1 if child is first live-born child to mother; 0 otherwise
*INTLT*16	≡	1 if current delivery is not the first, and previous delivery was less than 16 months earlier; 0 otherwise
*EDLT*12	≡	1 if mother did not finish high school; 0 otherwise
SMOKER	≡	1 if mother indicated she currently smokes, as of the October 1973 interview; 0 if mother indicated she never smoked or had stopped smoking sometime in the past; .43 if mother did not respond to the smoking question in the October 1973 interview[a]
SUCCESS	≡	Number of previous live births to mother divided by number of previous pregnancies
Y_n, N: Nonwage income and family size		
NEARNPC	≡	Family monthly nonearned income at pre-enrollment (measured in hundreds of dollars) divided by pre-enrollment family size
W: Wage rates		
PWAGEMOM	≡	Predicted hourly market wage rate of mother
ERNOTHPC	≡	Earnings of household members other than the mother at pre-enrollment (measured in hundreds of dollars) divided by pre-enrollment family size
t, G: Treatment parameters		
TREAT	≡	1 if family is in any of the four NIT plans; 0 otherwise
*AGELT*18*XT*	≡	1 if mother was younger than 18 at the time of child's birth and mother is in any of the four NIT plans (i.e., *AGELT*18 = *TREAT* = 1); 0 otherwise
*AGEGT*34*XT*	≡	1 if mother was more than 34 years old at time of child's birth and mother is in any of the four NIT plans (i.e., *AGEGT*34 = *TREAT* = 1); 0 otherwise
*INTLT*16*XT*	≡	1 if current delivery is not the first, previous delivery was less than 16 months earlier, and mother is in any four of the NIT plans (i.e., *INTLT*16 = *TREAT* = 1); 0 otherwise
SMOKEXT	≡	1 if mother indicated she currently smokes, as of the October 1973 interview, and mother is in any of the

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TABLE 5 (*continued*)

		four NIT plans (i.e., *SMOKER* = *TREAT* = 1); 0 if mother indicated she never smoked, had stopped smoking some time in the past, or was in a control family; .43 if mother did not respond to the smoking question in the October 1973 interview and mother is in any of the four NIT plans
Stratification variables		
AFDCMED	≡	1 if eligible for Medicaid; 0 otherwise
MOMHEAD	≡	1 if mother is married with spouse present and her husband is the household head, or mother is household head; 0 otherwise

a 139 missing observations on the smoking variable were recoded to .4302, the mean value for those who did respond to the smoking question. See Kmenta [20], p. 340, on techniques for recoding missing observations.

TREAT is a binary variable indicating experimental status. It is used instead of variables describing the experimental tax rate and guarantee (t and $G(N)$) for pragmatic reasons. Both experimental and control families may participate in a variety of other public welfare programs, each with its own tax rate and guarantee. Moreover, experimental families in the Gary Income Maintenance Experiment were guaranteed a minimum payment equal to what they would be eligible to receive under AFDC. Hence, the experiment's t and G separately do not distinguish adequately between an experimental effect and the economic environment in the absence of an income maintenance program. The variables *AGELT*18*XT*, *AGEGT*34*XT*, *INTLT*16*XT*, and *SMOKEXT* are interaction terms which disaggregate the treatment effect by the likelihood of an adverse pregnancy outcome. Since the medical and public health literature suggests that adverse pregnancy outcomes are associated particularly with mothers younger than 18 and older than 34, short interval between pregnancies, and smoking, these characteristics were used to define the treatment interaction terms.

Two variables in Table 5 are included to capture systematic differences in characteristics that may be associated with variations in birth weight across families. Eligibility for Medicaid (*AFDCMED*) represents systematic differences in families' budget sets, since persons eligible for Medicaid face a lower market price of medical care than do noneligible persons. It is expected that Medicaid eligibility will be associated with greater utilization

which household members have influence on the opportunity set (or resource constraint) of a given expectant mother.

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of prenatal care and, hence, with a lower incidence of low birth weight. The binary variable, *MOMHEAD,* indicates the mother's role in the family and is used to control for variations in the extent to which family resources may be available for use by the mother and variations in the labor supply response of expectant mothers to the wage rates of other family members. A priori, we expect at a minimum that the situations in which the expectant mother is either the household head herself or married to the head would be the ones in which she has the greatest claim on family resources.

Finally, it should be noted that several of the explanatory variables enumerated in this section are based on pre-enrollment data. These variables are the predicted market wage rate for the mother, monthly earnings of family members other than the mother, and family monthly nonwage income. Observations on these variables current to the pregnancy are likely to contain endogenous responses to the experiment. Consequently, it is preferable to use pre-enrollment observations for these variables (Moffitt [25]).[13]

IV. FINDINGS

Data

The data used in this study come from the data files assembled by the Gary experiment and from the birth records of children born to experiment participants. The experiment's data files are described in detail elsewhere (Kehrer et al. [17]). In this section, we focus on the procedures used to identify births and on the process of birth records data collection.

The infants born during the experiment were identified by generating a list of all persons in participant families born after December 31, 1970. There are two major sources for this information. First, all participating household units were required to submit monthly income report forms. In addition to data on income, this form called for information on changes in family composition, including the name, sex, and date of birth for each person who entered or left the family and the reason for the change. Also, at

13 As indicated in fn. 3, treatment status (i.e., treatment or control) is not purely random, but only random conditional on characteristics of participant families. As a consequence, Barnow, Cain, and Goldberger [2] point out, if "there is some characteristic of the treatment (or control) group that is both associated with receipt of the treatment and associated with the outcome [under study]," selectivity bias may arise. In our study, the variables that influence assignment to treatment status are included among the exogenous variables in the estimated reduced-form equation. Furthermore, (unlike Barnow-Cain-Goldberger) there are no *unobserved* variables commonly influencing the outcome and treatment status. Therefore, if we make the same distributional assumptions as Barnow-Cain-Goldberger, there is no selectivity bias.

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each of the eight regular interviews during the course of the experiment, the family head was asked to enumerate the persons in the household at the date of the interview.

A total of 697 names were identified from these sources. For these infants, birth records on file at the Indiana State Board of Health were then sought in an attempt to obtain information on birth weight, mother's previous obstetric history, and plurality. Of the 679 birth records sought, 612 were found. Given the proximity of Chicago as a potential place to have a child and the possibility of going to other states for the delivery, the 90 percent retrieval rate appears reasonably successful. There is no reason to expect any association between experimental treatment status and whether a child's birth record was found.

After inspecting the retrieved records, 148 were rejected from the study. These rejections were usually for one of the following reasons:

- The child was born before its family was enrolled in the experiment.
- The child was born after enrollment, but the mother's third trimester of pregnancy had begun before enrollment.
- The child's mother was not a member of the household, or had refused to participate in the experiment.
- It was not possible to determine which woman in the household was the mother.[14]
- The child moved into the household after birth, and the mother appeared not to have been in the family during her pregnancy.
- The mother moved into the household after the third trimester of pregnancy had begun.
- Birth weight was not recorded on the birth record.

Sixty additional cases were excluded because of plural birth or missing observations on explanatory variables. The final analysis sample consisted of 404 single live births, of which 256 belonged to experimental treatment families and 148 to controls. Mean birth weight for treatment infants was 3147 grams (standard deviation, 517 grams), while for control infants it was 3173 grams (standard deviation, 548 grams).

Estimated Response

Interpretation of the estimated coefficients in a Tobit model is more complex than in a linear regression model. In particular, the response of the dependent variable to a unit change in a given independent variable is not given by the estimated coefficient for that independent variable. (For binary

14 Restrictions on data availability imposed by the Indiana State Board of Health included the mother's name. In most cases, however, an infant's mother was known from the experiment's records or her identity could be inferred by comparing data on the birth record (e.g., mother's age and education) with data in the experiment's files.

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regressors, even the sign of the estimated coefficient does not necessarily describe the direction of the response.) Rather, the response depends upon the particular values assumed by all the independent variables and their estimated coefficients.[15] Hence, the impact of a change in one (or more) independent variable(s) can be assessed only for some specified set of values for the other independent variables. For this reason, in this section we present the estimates of birth-weight response to the income maintenance plan. The estimated coefficients along with the sample means for the explanatory variables are given in Appendix Table A-1.

The estimated response of birth weight to an NIT is presented in Table 6 for 12 cohorts of mothers. These 12 cohorts are defined by varying the combinations of treatment status and the following risk factors: mother's age (under 18, 18 to 34, or over 34); whether she smokes; and whether there was a short interval between pregnancies. Independent variables other than the treatment binary (*TREAT*), the risk factors of interest (*AGELT*18, *AGEGT*34, *SMOKER,* and *INTLT*16), and the risk factor–treatment interaction terms (*AGELT*18*XT, AGEGT*34*XT, SMOKEXT,* and *INTLT*16*XT*) are evaluated at their sample means. Standard errors are given in parentheses.

The estimated response is presented in the last two columns of Table 6, in grams and in pounds, respectively.[16] For each cohort, the gross effect $[E^T(R) - E^C(R)]$, where T refers to treatment observations and C to controls, is the difference attributable to income maintenance in the means of the deviation of birth weight from the optimal range. Note that a *negative* sign on an estimated effect represents a *reduction* in the deviation and should be interpreted as a *beneficial* effect on birth weight. Conversely, a *positive* sign on an estimated effect indicates that the deviation is *wider* among experimentals than controls, and hence should be interpreted as an *adverse* effect.[17]

The estimated effects show a quite consistent pattern, though they do not uniformly meet our a priori expectations. In nine of the twelve cohorts, there is a beneficial treatment effect, ranging between +0.3 and +1.2 pounds, and in four cohorts the effect is statistically significant at the 5 percent level. Moreover, these four are very likely the highest risk cohorts:

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15 See Kehrer and Wolin [16] for a more detailed exposition of the estimated response calculation.

16 For comparison, we also estimated the model with a dependent variable using 2501 as the critical value α; the estimated response based on this formulation is shown in Appendix Table A-2. As expected, the results do not differ appreciably, though the estimated responses are somewhat smaller.

17 Multiplying $[E^T(R) - E^C(R)]$ by minus one gives the average weight gain (or loss) associated with experimental treatment for infants whose birth weight would have been below the optimal range in the absence of income maintenance.

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TABLE 6
ESTIMATED NIT TREATMENT EFFECTS BY MOTHER'S AGE, INTERVAL BETWEEN PREGNANCIES, AND SMOKING STATUS

Cohort				Estimated Effects $E^T(R) - E^C(R)$ (Standard Error)	
Number	Age	Interval < 16 Months?	Smoker?	In Grams	In Pounds[a]
1	<18	Yes	Yes	−530.4**	−1.17**
				(185.2)	(0.41)
2	<18	No	Yes	−174.1*	−0.38*
				(86.2)	(0.19)
3	<18	Yes	No	−189.2	−0.42
				(129.6)	(0.29)
4	<18	No	No	12.2	0.03
				(42.0)	(0.09)
5	18–34	Yes	Yes	−282.5*	−0.62*
				(132.6)	(0.29)
6	18–34	No	Yes	−4.2	−0.01
				(56.2)	(0.12)
7	18–34	Yes	No	−6.9	−0.02
				(91.8)	(0.20)
8	18–34	No	No	117.6**	0.26**
				(33.3)	(0.07)
9	>34	Yes	Yes	−470.5*	−1.04*
				(210.6)	(0.46)
10	>34	No	Yes	−119.3	−0.26
				(110.4)	(0.24)
11	>34	Yes	No	−130.1	−0.29
				(150.2)	(0.33)
12	>34	No	No	65.8	0.14
				(70.1)	(0.15)

a Estimated effect in grams ÷ 453.6 grams.
* Estimated effect is statistically significant at the .05 level.
** Estimated effect is statistically significant at the .01 level.

mothers in each of the three age groups who both smoke and had a short interval between pregnancies, and mothers under age 18 who smoke.

Three cohorts in Table 6 (which very likely are among the lowest risk categories) show an adverse response to income maintenance. In two of the three (mothers under 18 and over 34 who neither smoked nor had a short interval between pregnancies), the response is not distinguishable from zero at the 5 percent significance level. The adverse treatment effect (a loss of

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0.26 pound) for the lowest risk cohort, mothers between 18 and 34 who neither smoked nor had a short interval between pregnancies, is statistically significant at the 1 percent level. This apparently perverse impact of income maintenance on the healthiest cohort is troublesome. One possible explanation for this finding is the method of sample selection. Because birth records were collected only for infants recorded in the experiment's files as entering families, some infants who died within a month of birth probably were never listed as having entered the family. Most of these children very likely had low birth weights. If the low-weight babies whose births were not recorded were concentrated among mothers in control families, the sample would understate the effect of income supplements on the reduction in the incidence of low birth weight. Such differential reporting between experimentals and controls is plausible because transfer payments of experimentals were a function of family size. Consequently, treatment families had a greater incentive than control families to report new family members.[18]

Response by Risk Class

Many of the cohorts shown in Table 6 can be ranked pairwise by risk of low birth weight. For example, an infant born within 16 months of a previous delivery to a mother under age 18 who smokes bears a higher risk of having low birth weight than an infant whose mother is under 18 but neither smokes nor has a short interval between pregnancies. Given the 12 cohorts in Table 6, 66 comparison pairs are possible. Of these, we feel competent to infer which of the two cohorts in a pair faces a greater likelihood of low birth weight for 23 of the 66 pairs. Our judgments were based on the following principles:

- The age group 18 to 34 is lower risk than either the age group under 18 or the age group over 34, other things equal.
- Expectant mothers who either smoke *or* have a short interval between pregnancies are lower risk than expectant mothers who have *both* these characteristics, other things equal.
- Expectant mothers who neither smoke nor have a short interval between pregnancies are lower risk than expectant mothers who have either of these characteristics, other things equal.

Table 7 shows the 23 comparison pairs of low-birth-weight risk along with a comparison of the estimated NIT effects for each pair.

For all 23 pairs, higher risk pregnancies uniformly show larger responses to treatment. Holding constant the two risk characteristics other

18 In this regard, note that the number of births in our sample relative to the number of families in the experiment is higher for experimentals than for controls (.25 vs .19). Although it is conceivable that this result is due to effects of the experiment on fertility, we believe the result is more likely to reflect differential reporting of births.

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TABLE 7
COMPARISON OF ADVERSE PREGNANCY OUTCOME RISK AND BIRTH-WEIGHT RESPONSE TO NIT TREATMENT FOR 23 PAIRS OF COHORTS OF MOTHERS[a]

Cohorts	Risk Comparison	Effect Comparison	Cohorts	Risk Comparison	Effect Comparison
1, 2	1 > 2	1 > 2	5, 9	9 > 5	9 > 5
1, 3	1 > 3	1 > 3	6, 8	6 > 8	6 > 8
1, 4	1 > 4	1 > 4	6, 10	10 > 6	10 > 6
1, 5	1 > 5	1 > 5	7, 8	7 > 8	7 > 8
2, 4	2 > 4	2 > 4	7, 11	11 > 7	11 > 7
2, 6	2 > 6	2 > 6	8, 12	12 > 8	12 > 8
3, 4	3 > 4	3 > 4	9, 10	9 > 10	9 > 10
3, 7	3 > 7	3 > 7	9, 11	9 > 11	9 > 11
4, 8	4 > 8	4 > 8	9, 12	9 > 12	9 > 12
5, 6	5 > 6	5 > 6	10, 12	10 > 12	10 > 12
5, 7	5 > 7	5 > 7	11, 12	11 > 12	11 > 12
5, 8	5 > 8	5 > 8			

a Cohort numbers are given in Table 6

than mother's age, mothers under 18 show a greater average response than mothers between 18 and 34. Also, mothers over 34 show a stronger response than the 18- to 34-year-old cohorts. Comparisons within age groups show a similar consistency. Mothers who smoke and have children in quick succession always show a larger estimated response than other groups within the same age cohorts. On the other hand, mothers who neither smoke nor have a short interval between deliveries always show the smallest response to the NIT for their age cohorts.

Subsidiary Findings: The Intervening Process

We speculated above that income maintenance might reduce the incidence of low birth weight through increased use of prenatal care, improved nutrition, or a reduction in hours worked at paid employment. In view of our finding that experimental NIT treatment status was associated with a lower incidence of suboptimal birth weight among children born to women with high-risk pregnancies, it is instructive to consider which, if any, of the hypothesized factors contributed to the observed effects. It seems unlikely that reduction in hours worked was a significant factor. About 35 percent of the mothers in our study sample were teenagers and hence probably were

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not employed prior to their pregnancies. In addition, labor force participation rates were relatively low among the adult women in the Gary experiment: only about 15 percent of wives and about 20 percent of female family heads were employed.[19] Some limited exploratory analysis regarding the possible influences of prenatal care and maternal nutrition was conducted, however, and the results are presented below.

Prenatal Care. Indiana's Certificate of Live Birth contains two items concerning prenatal care: the month in which care began, and the number of prenatal physician visits. Of the 404 observations in our analysis, 172 certificates reported the month prenatal care began and 179 had entries for the number of prenatal visits. If prenatal care was part of the causal mechanism for improvement of newborn health status, the distributions of those prenatal care variables should differ by NIT treatment status. In particular, experimental groups who showed a positive response should have used more and earlier prenatal care.

The distributions of the prenatal care variables, shown in Table 8, are similar for treatment and control observations.[20] These is no clear pattern of either more visits or earlier utilization for treatment women. In addition, there is no systematic relationship between treatment status and nonreporting on the prenatal care utilization variables.[21] On the basis of this sketchy evidence, we cannot attribute the observed birth-weight response to patterns of health-care utilization. We are hesitant to dismiss the argument that prenatal care had no influence because we have no data on types or quality of care used. However, the evidence at hand weakens the proposition that the mechanism underlying the birth-weight response was physician care.

Nutrition. The Gary experiment did not collect data on participants' diet, so no direct evidence is available that would permit inference about whether the observed birth-weight response is attributable to improvements in nutrition. However, some indirect evidence is available. First, nutrition data were collected as part of the Rural NIT experiment. Analysis of these data suggested that "the negative income tax is associated with a small but statistically significant improvement in the quality of diets of families in

19 Although the experiment led to a moderate decrease in hours worked for female family heads, it did not lead to any such decrease for wives.

20 The data shown are disaggregated only by treatment status. We examined the data disaggregated further by the risk cohorts defined in Table 6. However, the pattern of distributions did not change. Hence, those results are not shown, but are available from the authors.

21 The number of physician visits was available for 42 percent of the controls and 46 percent of the treatment women. Similarly, the birth records for 41 and 44 percent of the control and treatment observations, respectively, contained information on the timing of the first prenatal visit.

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TABLE 8
FREQUENCY DISTRIBUTION OF NUMBER OF PRENATAL PHYSICIAN VISITS AND MONTH OF PREGNANCY PRENATAL CARE BEGAN, BY NIT TREATMENT STATUS
(Percent)

Treatment Status	Number of Prenatal Visits					
	0	1–3	4–6	7–9	10–12	13+
Treatment (N = 117)	0.9	12.8	32.4	22.2	24.7	7.0
Control (N =62)[a]	0.0	11.3	32.2	25.8	24.1	6.4

Treatment Status	Month of Pregnancy Prenatal Care Began[b]		
	1–3	4–6	7–9
Treatment (N = 112)	49.2	49.1	1.8
Control (N = 60)	50.0	45.1	4.0

a Total does not add to 100.0 because of rounding error
b Of those who had at least one visit.

North Carolina, but not in Iowa" (O'Connor, Madden, and Prindle [35]).[22] These results lend some support to the plausibility of maternal nutrition as a mechanism influencing infant health, though the populations in the Gary and Rural experiments are sufficiently different that the findings from one experiment cannot be assumed to characterize the other.

As additional indirect evidence bearing on the plausibility of nutrition as a source of the birth-weight response, we considered whether treatment-control income differences were likely to be sufficient to provide significant improvement in the nutrition of pregnant women. A rough measure of the income required for adequate nutrition was developed from published data and compared to income differentials of experiment participants.

The differences between estimated monthly expenditures on food during a representative calendar quarter (the first three months of 1974)[23] and expenditures required for a "typical" low-income family with one pregnant female were calculated and compared for treatment and control families. On average, families of three or more spend about one-third of their income on food (Orshansky [36]).[24] During the first quarter of 1974,

22 See also another version of this paper, "The Negative Income Tax and the Quality of Dietary Intake," pages 507–17, in this issue.
23 Income for 1974 is used for our calculations since published data on food expenditures required to meet nutrition standards are in 1974 prices (Peterkin [38]).
24 Naturally it is preferable to use an Engel curve (for each family size) to estimate expenditures on food. However, available data on food expenditures as a percentage of income by

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the average monthly income for a control family of four in our analysis sample was about $296. Such a family would have spent about $98 per month on food. In contrast, an average treatment family of four had a monthly income of about $453 and would have spent $150 on food.

Neither expenditure level is, on average, adequate to meet the 1974 Recommended Daily Allowance (RDA) adopted by the U.S. Department of Agriculture (USDA). The USDA, using data on food consumption for families in the 26th to 49th percentile of money value of food per person in a week (Peterkin [38]) estimates that food expenditures of $197.80 per month would be necessary for a family of four (one male adult, one female adult, and two school-aged children) to meet the RDA. When the adult female in the "typical" family is pregnant, the standard rises by $11.00 to $208.80 per month—implying a food expenditure deficit of $110.80 for the typical control family of four, but a deficit of only $58.80 for the typical treatment family of the same size. (Similar calculations for families of five indicate a deficit of about $41.00 for treatments and $106.00 for controls.) Thus, the higher incomes of treatment families would have permitted an improvement in the nutrition of their members.[25] These results are consistent with the argument that improved nutrition was part of the mechanism responsible for the observed birth-weight response to NIT treatment in Gary.

V. CONCLUSION

The results reported in this study suggest that children who would have had suboptimal birth weight and whose mothers were at particularly high risk of experiencing an adverse pregnancy outcome will exhibit an average birth-weight gain of about 0.3 to 1.2 pounds in response to NIT treatment at the level provided in the Gary experiment. Supplemental exploratory investigation suggests that improvements in nutrition may have been part of the causal mechanism responsible for the observed response. Additional research now in progress will seek to replicate these findings for a sample of participants in the Seattle/Denver Income Maintenance Experiment.

From the perspective of evaluating an NIT as a public policy instrument, two observations seem pertinent on the basis of the findings presented in this paper. First, the magnitude of the birth-weight response appears to be large enough to constitute a significant health effect. Rush et al. [40] argue

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family size (BLS [4]) are based on a small sample and suggest, implausibly, that food is an inferior good over the low-income range. Consequently we relied on the 33 percent figure used to calculate official poverty levels.

25 Note that some portion of the higher mean incomes of treatment families (holding family size constant) may reflect pre-enrollment differences. Hence, not all the observed income differential can be attributed to NIT treatment.

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that a difference in mean birth weight of .28 pound (125 grams) would constitute a significant improvement in infant health, since this difference is approximately the same as "the difference in birth weight between the offspring of cigarette smokers and nonsmokers, and . . . the differences in birth weight between American blacks and whites of similar social status." Judging by this criterion, an NIT plan will effect a significant health response among children of women who face high-risk pregnancies.[26]

Also, of particular interest in evaluating the efficacy of an NIT, the observed beneficial health effect resulted without the provision of direct health services. While it would not be appropriate to conclude that an NIT program is more efficient than a program of direct services in achieving beneficial health outcomes, the results presented here are evidence that an NIT may have a broad influence on social behavior and the quality of life.

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26 Note that Rush et al. [40] were concerned with a change in the mean, and our analysis focused on the left-hand tail of the birth-weight distribution. While a given change in the mean is likely to be associated with a larger change in the tail, the magnitudes of our statistically significant estimated effects are sufficiently greater than the 125-gram standard to warrant comparison.

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TABLE A-1
ESTIMATED NORMALIZED COEFFICIENTS,[a] STANDARD ERRORS, AND SAMPLE MEANS, REDUCED-FORM BIRTH-WEIGHT EQUATION

Variable	Estimated Normalized Coefficient (Standard Error)	Sample Mean
MALE	−0.1518	0.50
	(0.1244)	
*AGELT*18	0.4070	0.24
	(0.3028)	
*AGEGT*34	0.3954	0.11
	(0.3186)	
PARITY	0.0175	2.44
	(0.0384)	
PRIMPARA	0.6156	0.29
	(0.2724)	
*INTLT*16	0.9020	0.12
	(0.3008)	
*EDLT*12	0.1090	0.65
	(0.1524)	
SMOKER	0.8197	0.44
	(0.2664)	

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TABLE A-1 (*Continued*)

Variable	Estimated Normalized Coefficient (Standard Error)	Sample Mean
SUCCESS	−0.5146	0.94
	(0.3571)	
NEARNPC	−0.0248	0.14
	(0.2613)	
PWAGEMOM	−0.2774	2.24
	(0.1765)	
ERNOTHPC	−0.2181	0.32
	(0.1587)	
TREAT	0.8168	0.63
	(0.2422)	
AGELT18XT	−0.7345	0.15
	(0.3060)	
AGEGT34XT	−0.4398	0.05
	(0.4029)	
INTLT16XT	−0.8458	0.08
	(0.3754)	
SMOKEXT	−0.8356	0.30
	(0.3176)	
AFDCMED	−0.0914	0.70
	(0.1878)	
MOMHEAD	0.3072	0.64
	(0.2727)	
CONSTANT	−0.2583	1.00
	(0.6223)	

a Normalized coefficients are $\beta_1/\sigma, \beta_2/\sigma, \ldots, \beta_k/\sigma, 1/\sigma$, where σ is the standard deviation of the error term.

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TABLE A-2
ESTIMATED NIT TREATMENT EFFECTS BY MOTHER'S RISK COHORT, DEPENDENT VARIABLE BASED ON CRITICAL VALUE α = 2501 GRAMS

Cohort				Estimated $E^T(R) - E^C(R)$ (Standard Error)	
Number	Age	Interval <16 Months?	Smoker?	In Grams	In Pounds[a]
1	<18	Yes	Yes	465.8**	1.03**
				(224.8)	(0.50)
2	<18	No	Yes	175.3*	0.39*
				(90.4)	(0.20)
3	<18	Yes	No	71.4	0.16
				(67.4)	(0.15)
4	<18	No	No	8.2	0.02
				(10.7)	(0.02)
5	18–34	Yes	Yes	135.0	0.30
				(99.8)	(0.22)
6	18–34	No	Yes	1.4	0.00
				(30.5)	(0.07)
7	18–34	Yes	No	−22.0	−0.05
				(29.4)	(0.06)
8	18–34	No	No	−28.3**	−0.06**
				(12.2)	(0.03)
9	>34	Yes	Yes	389.8*	0.86*
				(221.5)	(0.49)
10	>34	No	Yes	131.1	0.29
				(86.0)	(0.19)
11	>34	Yes	No	47.0	0.10
				(58.6)	(0.13)
12	>34	No	No	0.3	0.00
				(13.0)	(0.03)

a Estimated effect in grams ÷ 453.6 grams.
* Estimated effect is statistically significant at the .10 level
** Estimated effect is statistically significant at the .05 level.

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